

1 Chapter 1

All:1.1,1.2,1.7,1.14 and 6 examples

1.1:

$$\begin{aligned}x &= \tan\theta d \quad \dot{x} = \frac{\dot{\theta}}{\cos^2\theta} d = \frac{\omega}{\cos^2\theta} d \\ \cos\theta &= \frac{d}{\sqrt{d^2 + x^2}} \Rightarrow \frac{\omega(d^2 + x^2)}{d} \\ \ddot{x} &= \frac{2x\dot{x}\omega}{d} = \frac{2x\omega}{d} \frac{\omega(d^2 + x^2)}{d} = \frac{2x\omega^2(d^2 + x^2)}{d^2}\end{aligned}$$

1.2:

$$\begin{aligned}\begin{cases} x = \sin\theta b \\ y = \cos\theta d \end{cases} \\ \frac{d}{dt} \cos\theta(b+d) &= -c \\ \Rightarrow \dot{\theta} &= \frac{c}{\sin\theta(b+d)} \\ \begin{cases} \dot{x} = \frac{cb\cos\theta}{\sin\theta(b+d)} \\ \dot{y} = -\frac{dc}{(b+d)} \end{cases} \\ \begin{cases} \ddot{x} = -\frac{c^2 b}{\sin^3\theta(b+d)^2} \\ \ddot{y} = 0 \end{cases}\end{aligned}$$

1.7:

$$\begin{aligned}\vec{v} &= v_x \vec{i} + v_y \vec{j} \quad v_x^2 + v_y^2 = c \\ \vec{a} &= \dot{v}_x \vec{i} + \dot{v}_y \vec{j} \\ \vec{a} \cdot \vec{v} &= v_x \dot{v}_x + v_y \dot{v}_y = \frac{1}{2}(\dot{v}_x^2 + \dot{v}_y^2) = \frac{1}{2}\dot{c} = 0\end{aligned}$$

1.14:

$$\begin{cases} m\ddot{x} = -mk\dot{x} \\ m\ddot{y} = -mk\dot{y} - mg \end{cases}$$

$$\begin{cases} \dot{x} = C_1 e^{-kt} \\ \dot{y} = C_2 e^{-kt} - \frac{g}{k} \end{cases}$$

$$t = 0 \quad C_1 = v_{x0}; C_2 = \frac{g}{k} + v_{y0}$$

$$- \tan\alpha = \frac{(\frac{g}{k} + v_{y0})e^{-kt} - \frac{g}{k}}{v_{x0}e^{-kt}}$$

$$- \tan\alpha = (\frac{g}{kv_{x0}} + \tan\alpha) - \frac{g}{kv_{x0}}e^{kt}$$

$$\Rightarrow e^{kt} = \frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2\tan\alpha)$$

$$t = \frac{1}{k} \ln[\frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2\tan\alpha)] = \frac{1}{k} \ln[1 + 2\frac{k}{g} \sin\alpha v_0]$$